(a) Tactile description

Paired fingertip signals · last T frames

P2L Left tactile clip

P2R Right tactile clip

t − T + 1 … t

Shared spatial CNN

Temporal convolution

Discrete tactile text

area · Fx · Fy · Fz · rotation

Reference: left-finger sensor frame

(b) Shared reasoning and action model

Tactile semantics

C Common inputs

P1F Front

P1W Wrist

Current 7D qpos

Task instruction

T Tactile text

Current Touch

H Motion history

qpos H100 → 11

P Current plan

none if unavailable

F Global failures

Current + past failures

Shared $\pi_{0.5}$ / PaliGemma backbone

Hold & reason

C T H P

Failure diagnosis

Global failure sequence

C F

Recovery planning

true

plan

Assessment lane

Execution

C T H P

need_recovery

false: continue current phase

Adjustment

C T H P

adjustment_end

adj = true

asynchronous

Action lane

Action Expert

execution C

adjustment C P

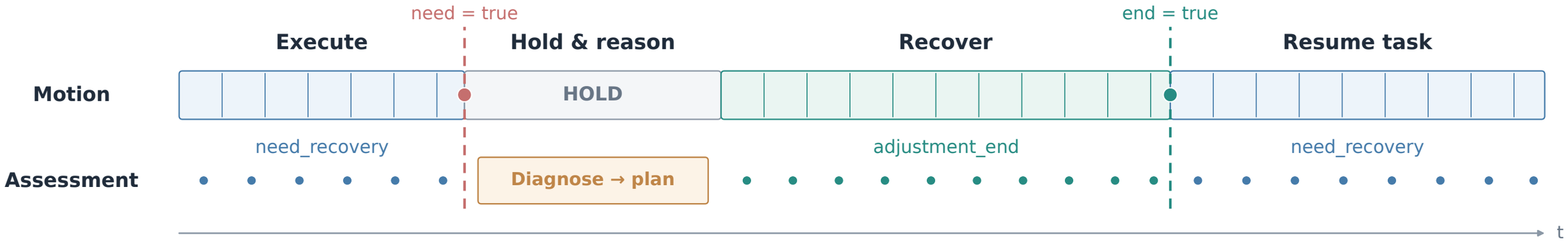
Flow matching

H30

P3 Robot motion

(c) Asynchronous execution–recovery loop

Monitor during motion · interrupt within a chunk



Resume, re-evaluate, and recover again if a fault remains.

Solid: information / action flow Dashed: feedback / stage control • Shared-model calls are serialized; monitoring overlaps physical motion.